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FINAL REPORT:

ANALYSIS OF THE RELATIONSHIP BETWEEN HEALTHY EATING AND MULTIPLE DISEASES

Subject: Proyecto II

Grado Ciencia de Datos, 2ºB2

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1. Scope of the project

Using datasets that provide data on GDP per capita, caloric intake, food consumption, prevalence of cardiovascular disease, as well as life expectancy, this project seeks to find some relationship between healthy eating and socioeconomic factors.

Requirements:

- ★ Express the tasks to be fulfilled to accomplish the project's objective.

Restrictions:

- ★ Lack of knowledge in R
- ★ Datos ausentes de

Project deliverables:

- ★ Hito 1
- ★ Hito 2
- ★ Final report
- ★ Powerpoint presentation

Project boundaries (what we will and will not do):

The analysis is not feasible to all countries as not all datasets share the same ones.

It should be taken into account aspects such as:

- ★ **Dates**: The dates on which the data was extracted and its relationship to the conclusions.
- ★ **Efficiency**: Fulfillment within the deadlines and use of the available tools.

1.1. Objectives of the project

Objective 1: To evaluate the influence of the level of economic development of countries on the eating habits of their citizens.

Objective 2: To investigate the relationship between the prevalence of diabetes and obesity levels across countries.

Objective 3: To analyze the relationship between life expectancy, dietary patterns, and cardiovascular health in American countries during years 2010-2014 .

Objective 4: To analyze the relationship between the prevalence of obesity and calorie intake across countries depending on Human Development Index (HDI) levels.

Objective 5: To analyze the relationship between healthcare expenditure and the prevalence of nutrition-related chronic diseases, such as obesity, hypertension, and cardiovascular conditions.

Objective 6: Do alcohol consumption, physical inactivity, or drug use influence obesity and diabetes?

1.2. Usefulness of the study

This study analyzes the relationship between economics, obesity and eating habits using data sets organized by country. Its main focus is to identify patterns and connections that allow relevant conclusions to be drawn, such as the possibility of predicting indicators like life expectancy or obesity levels. Another example of what can be obtained is information on the impact of different socioeconomic and health factors, such as GDP or diabetes prevalence, on eating habits in different countries. This is very useful for governments and for the population itself as it can be used to create preventive strategies on diet-related diseases or to raise awareness of the risks of poor diet and how to prevent them. In addition, this study could be suitable for people who change their residence to learn about the eating habits of the population of the country of interest.

Also, it can be useful for pharmaceutical companies that need to know which

geographical areas need more or less of a drug, for example ozempic.

Finally, it can be very useful to obtain key information to design public policies and health strategies focused on improving nutrition and reducing chronic diseases. Moreover, the analysis of dietary patterns in different socioeconomic contexts can help to identify which habits are healthier and which interventions are needed in less economically developed countries.

2. Configuration of the project

2.1. Data sources

A search has been carried out on different online platforms offering thousands of datasets, mainly Kaggle, but also Statista, the World Health Organization (WHO) website and Eurostat have been searched, which can be seen in Hito I (Annex 5.1.) After the search and discarding of several not complete datasets, the following datasets have been chosen to carry out the research:

GDP per capita

Obesity prevalence in the adult population

The Burden of Cardiovascular Diseases - PAHO/WHO | Pan American Organization

Hypertension - PAHO/WHO - Pan-American Organization

Life Expectancy (WHO)

Daily supply of calories per capita

Diabetes mellitus by country

Intake by food group

Prevalence of insufficient physical activity among adults aged 18+ years (age-standardized estimate) (%)

VizHub - GBD Results

Firstly, an exhaustive search was carried out on several platforms with a multitude of datasets, mainly kaggle, but also statista and the World Health Organization (WHO) have

been consulted. After having decided to carry out the investigation on nutrition, the most complete datasets providing the necessary information were from both WHO and Eurostat.

Several techniques were used in order to obtain the data, depending on the source and format available. The files in .csv and .xlsx formats were downloaded from official URLs, such as Eurostat and DBnomics, to obtain data on the prevalence of obesity in the adult population, among others.

In addition, other datasets such as 'GDP' were excluded because more information on this area could be found in another more complete dataset, 'Life Expectancy' to be exact. Another reason for its exclusion was the large amount of missing data contained in the 'GDP' dataset, its poor compatibility, in terms of countries, with the rest of datasets.

Also, two data sets on drug use and alcohol consumption have also been found in a reliable source such as the Institute for Health Metric Evaluation (IHME).

Furthermore, another research topic was discarded, regarding mental illnesses, for which a dataset was found that was disused due to its lack of relevance and lack of alignment with the project's objectives. Therefore, it was replaced by another one that provided information on cardiovascular diseases, much more useful for the study of the project.

2.2. Sources analysis

For data cleaning, a separate exploratory data analysis was performed:

OBESITY

The dataset includes data on the prevalence of obesity in both sexes from 1990 to 2021. This data is obtained taking into account individuals over 18 years old with a BMI of 30 kg/m^2 or higher

DIABETES

In this case, it covers data about the prevalence of diabetes by country, from both sexes, during the years 1980-2014.

LIFE EXPECTANCY

This dataset includes information about the life expectancy in various countries, from the year 2000 to 2015. There are multiple other variables, such as: adult mortality, alcohol consumption, IMC, thinness, PIB and health expenditure.

CALORIES

From 2000 to 2018. It includes a total of 3 different variables: countries, average daily kcal consumed per capita derived from animal foods per country, average daily kcal consumed per capita from plant foods per country and the average daily kcal consumed per capita per country. The variables can be very useful to know if there is a relationship with BMI and kcal intake or to know if GDP influences the caloric intake, so it can be highly useful for the study.

DIETARY INTAKE

It contains data by country, from 2010 to 2022, on food types, specifically cereals, fruits, vegetables, sugar, palm oil and animal fats; as well as its caloric quantity measured by daily capacity.

HYPERTENSION

It includes, by country and year, the prevalence of deaths due to hypertension, adding information about the gender of each group, male or female.

CARDIOVASCULAR DISEASES

In this dataset there is information, by country and year, on the prevalence of deaths due to cardiovascular diseases, according to gender, male or female.

ALCOHOL

This dataset consists of 4185 rows, which contain information on four variables, such as country, year, country code and the number of liters of pure alcohol consumed by people over 15 years of age per year.

DRUGS

With a total of 4334 observations this dataset has a total of 10 variables, all of which are related to the prevalence of substance use. The main variables to be used are country, year and value, the latter referring to the prevalence of drug use.

PHYSICAL INACTIVITY

This dataset, which contains 3690 rows, obtained through the WHO, collects data on the prevalence of people over 18 years of age in the countries studied who do not practice enough physical activity. It has multiple variables, the most important for this project are location, FactValueNumeric and period.

2.3. Data integration and transformation (reevaluar HITO 2)

This entire section is included in Hito II (Annex 5.2.)

Firstly, before carrying out any kind of transformation, missing values have to be treated. The amount of missing data was relatively big, however they were included in variables that were not relevant for each specific objective, so they were deleted. From the variables that are used in the study, there are only missing values in 'Life Expectancy'. In the dataset 'Life Expectancy' there were a total of 10 missing values, all of them stored in the variable 'adult mortality'. The missing values are not treated because, when merging 'Life Expectancy' with the other datasets for the objective 3, which is the only one that uses 'adult

mortality', the missing rows were automatically excluded, as they did not match the other datasets.

One of the challenges found in the transformation of data was the inconsistent formatting of some of the numeric variables. Several variables were stored using commas for decimal values (30,1 instead of 30.1), so they were being treated as categorical. To solve the issue, they were replaced by periods in R.

In the 'Calories' dataset, the process of category unification was performed. The 'Product' variable contained 9 variations of the main 3 categories (vegetal, animal and total). Additionally, in 'Pais' the same occurred, where instead of manually cleaning the imperfections, the function trimws() in R was used, because there were even more different categories.

Moreover, in 'Calories', each year is represented as a column, with its corresponding amount of calories as its data. In order to allow an easy integration with other datasets, the dataset is transformed so there is a single column with the years, and instead of having one 'Product' column, it is separated into three distinct columns, one for each calorie type (animal, vegetal, total), with their respective kcal values.

A similar process was performed to the 'Dietary Intake' table, where each unique value of 'Item' was transformed into different columns; and each data from the 'kcal.p.cap' column is now the value for each respective item.

Before merging the datasets, it is essential for the columns that are shared across tables to have the same name, as it is required for merging using R. For instance, in the table 'Dietary Intake', the variable under 'Area' contains all the countries, but it is changed to 'Pais' to match with its corresponding column in 'Calories'.

As for the datasets on drugs, alcohol and physical inactivity, no null data were found in any of them. However, at the time of joining them together with the diabetes and obesity datasets, by year and country, most of the data had null observations and therefore, when these rows were eliminated, the dataset had only 556, which was sufficient for the analysis performed.

Once the datasets were all cleaned, data integration was performed. The various datasets could not be fully merged because some countries did not match across all databases, so as a result, some datasets were intersected separately. With this, data consistency was ensured.

2.4. Tools and techniques used

The main tools and techniques used during the development of this project are:

- RStudio, which is the most suited environment to operate with the programming language R.
- The different libraries such as 'DataExplorer', ‘dplyr’, ‘ggplot2’,‘cluster’ and ‘FactoMineR’ have been used to perform the initial data analysis.
- K means Clustering (Elbow Method), Pearson/Spearman Correlation, Mapping, Silhouette Score, PCA,Correlation matrices,linear regression models…

3. Results

3.1. Objective 1

Three datasets—life expectancy, calories, and nutritional intake—have been combined to effectively meet the primary goal. This dataset compiles data from 30 European nations spanning 2010 and 2015. Furthermore, there are variables such as the proportion of calories from animal and plant sources, the ratio of calories from animal to plant sources, and a dietary diversity index, which offer a better analysis of dietary changes in relation to the economic development of countries.

First, a comparative study has been done to highlight the dietary discrepancies between wealthy and poor nations. Thus, box plots show that developed countries have diets richer in plant calories and animal fats, whilst developing nations seem to get a larger proportion of calories from animal sources and sugars. A bar chart is presented here showing

the influence of cultural and economic aspects on individual dietary choices. This indicates that it is essential to adjust the diet to the specific requirements of each social group (Annex 5.3.1).

In addition, when analysing the temporal evolution of these dietary habits, a slight increase in the amount of calories from animals can be seen in both groups. This increase is most noticeable in developing nations, suggesting a transformation in their diets, probably driven by economic growth and evolving tastes.Understanding these trends and linking them over time is crucial for foretelling future public health problems and modifying intervention approaches (Annex 5.3.2).

To better understand how dietary variables are connected, we have used Principal Component Analysis (PCA), which has simplified the data, reducing its complexity and pointing out the major influences explaining most of the variances in food habits across nations. This chart draws attention to nations with similar economic traits (Annex 5.3.3).

On this basis, we used hierarchical clustering with Ward's method on the standardised variables; the dendrogram revealed distinct and cohesive groups of countries matching their economic status, whether developed or developing, which supports the idea that dietary habits are influenced by economic development and validates the clusters proposed by the PCA (Annex 5.3.4).

At last, we used linear regression techniques to evaluate how dietary patterns are affected by socio-economic and health elements. With predictors such HDI, GDP per capita, BMI and year, we looked at the results for the dietary variety and percentage of calories from animal sources. Manifesting their major influence in influencing the food choices of the whole people, HDI and GDP are shown to be important variables ($p < 0.01$). Adding BMI as a control variable has therefore given the study more depth by looking at how health condition might influence food selections. Annex 5.3.5 offers a summary of the regression findings including the coefficients and their importance.

We have analyzed by means of a correlation matrix how dietary, economic, and health elements are related. Along with scatter plots showing the fitted regression lines, this study has found numerous significant connections that support past findings. Positive correlations between GDP per capita and the percentage of calories derived from plant sources, as well as

between the HDI and dietary variety, for instance, were discovered. On the other hand, adult mortality has demonstrated bad relationships with life expectancy and some specific dietary intakes, thereby helping one to more grasp how socioeconomic and health variables affect food habits (Annex 5.3.6 presents the correlation matrix and the most important scatter plots).

Ultimately, studies on people's eating habits reveal the major impact economic growth has on them. Generally speaking, richer nations eat more calories from animal fats and plant foods; conversely, developing nations have diets heavy in animal calories and sweets. Regression and correlation studies confirm and quantify these relationships, highlighting economic expansion and human development as significant drivers of dietary changes; multivariate and clustering techniques strengthen these trends and reveal their close relationship with a country's economic state. Finally, we would like to point out that these data offer insightful direction for creating nutrition programs customized to various economic situations; the graphic components in the study support these findings by plainly displaying the multifaceted and varied features of the data.

3.2. Objective 2

For this objective, the final dataset was created unifying 5 datasets: ‘Diabetes’, ‘Heart diseases’, ‘Obesity’, ‘Hypertension’ and ‘Life Expectancy’. Even though the objective only considers the prevalence of diabetes and obesity, other databases have been joined in order to also use the data to solve the objective 5. This was done for the dataset to have more consistency and to allow better comparisons. Unlike for the other merged dataset, this one only includes countries from the American continent. All variables were scaled, as they were in quite different units.

First, to achieve this objective, most of the variables were omitted, leaving only country, year, prevalence of diabetes and average obesity. Since only two numerical variables have been used, it has only been possible to create one, diabesity_index, which is a standardized variable that combines the values of obesity and diabetes prevalence. Once the other variables had been trimmed, the Pearson correlation index between diabetes and obesity was calculated. Thus obtaining a correlation of 0.734 and obtaining the scatter plot which shows a clearly positive relationship for all the countries in all the years studied (Annex 5.3.22.).

However, there are countries that have maintained a more stable evolution throughout the years studied (2000-2014) such as Canada. Other countries like Jamaica, Bahamas or Cuba show an evident growth in the development of diabetes and obesity (Annex 5.3.23.),(Annex 5.3.42.).

3.3. Objective 3

Furthermore, to give an answer to objective 3, three files needed to be combined: 'Life Expectancy', 'Cardiovascular diseases' and 'Dietary intake'. As with the previous merged dataset, all the countries included are from the Americas. The aim of this analysis is to study how dietary patterns and cardiovascular health are related to life expectancy during the period 2010-2014. Therefore, of the variables in the original dataset, we selected those that provided information on nutrition, cardiovascular health and other variables such as year or life expectancy in order to obtain a more appropriate analysis. Others have been discarded as they are not relevant or contain a single value. The variables studied can be seen in the Annex 5.3.41. and it can be observed that they do not have any missing data.

As for the diet variables, they measure the amount of daily calories per person (kcal/cap/d) by food groups. Furthermore, to deepen the analysis and better summarise dietary and cardiovascular patterns, two new variables derived from the original ones have been created: "Calorias.totales" (sum of dietary variables) and "Indice.riesgo.card"" (obtained by combining the prevalence of hypertension and heart disease, with prior standardisation).

A total of 200 observations, 19 different countries and 14 variables were obtained from which an exploratory analysis was carried out. It has been observed that the great majority of the variables present asymmetry, outliers or do not have a normal distribution, as we can see in annexes 5.3.30. and 5.3.31.

Therefore, the Spearman correlation method is chosen as it is more robust. From the correlation matrix (Annex 5.3.32.) it can be seen that there are several variables positively related to Esperanza de vida, such as 'Grasas animales' o 'Calorias.totales'. This may indicate that access to a high-calorie diet may be related to a high life expectancy. Among the health indicators we find a weak negative relationship with the prevalence of hypertension. And the strongest negative relationship is with adult mortality. These results are striking because of the relationship between animal fats, heart prevalence and life expectancy.

For a better analysis a regression model (Annex 5.3.33.) is used, using life expectancy as the dependent variable. The significant variables are grasas animales, frutas, azúcares y cereales. It confirmed that some of the relationships observed in the correlation matrix were statistically significant, such as Grasas Animales (Annex 5.3.34.).

As there are contradictory results (especially for animal fats) and considerable variability between countries, clustering is carried out to look for possible patterns between countries according to their dietary and cardiovascular patterns. For this purpose, different methods have been used to select the correct number of clusters. These have been the Nbclust (Annex 2.3.36.) using k-means and the elbow method (Annex 5.3.35.) using "wss". The result of this shows 3 clusters differentiated by dietary patterns and prevalences of cardiovascular diseases. In addition, each country appears in several rows for each year and sex. Thanks to this, it has been possible to study the temporal evolution of the countries belonging to each cluster and their changes (Annex 5.3.37.). The most frequent countries in each cluster have also been studied (Annex 5.3.38.). After this, an analysis of the distribution of the variables according to the cluster has been carried out (Annex 5.3.39. and 5.3.40.) where it can be observed:

- Cluster 1: stands out for its high consumption of animal fats and cereals compared to the others. The same for Fruits and Vegetables. In terms of health indicators, it has the lowest values for cardiac prevalence and hypertension. The same is true for sugars, where a moderate and lower consumption than the others is observed. Therefore, this cluster may represent a healthier profile, also finding that it is the cluster with the highest life expectancy and the largest female population. Some countries in this cluster are Canada or Argentina.

- Cluster 2: the countries in this cluster consume high levels of sugars and medium levels of cereals. They also have the highest prevalence of heart disease and hypertension and the lowest levels of animal fats. This may be related to the average life expectancy. These countries may have nutritional imbalances and a more male-dominated population. Countries such as Honduras or El Salvador.
- Cluster 3: It has the lowest consumption of Cereals, Vegetables and Cardiac prevalence. In terms of the other variables, it has intermediate levels. This cluster doesn't have the worst nutritional profile, but it does have the lowest life expectancy, this may be due to variables not studied in this objective. However, instability in countries like Jamaica or Costa Rica is notable.

3.4. Objective 4

Moreover, for the objective 4 its dataset is created from the combination of 3 files: 'Life Expectancy', 'Obesity', 'Calories'. The final set of data is composed of European countries, just like the second set, in specific, 25 countries.

Although there are 25 countries, there is information from 2000 to 2015 which makes 400 rows of data with variables that are going to be needed like Med.Obesidad, Total.Kcal, Animal.Kcal, Vegetal.Kcal, IDH, Life Expectancy, Thinness and Mortality.

This objective is focused on analyzing the relationship between the prevalence of obesity and calorie intake across countries depending on their Human Development Index (HDI) levels. As it had to be taken into account the HDI values of each country to realize the study, countries have been separated into two clusters (better silhouette coefficient) using IDH, Med.Obesidad and Total.Kcal. One variable was not enough to classify well in clusters, but either way, as it can be seen in the scatter plot in the Annex 5.3.7., IDH turned out to be the most relevant variable in the partition using K Means and data in 2015.

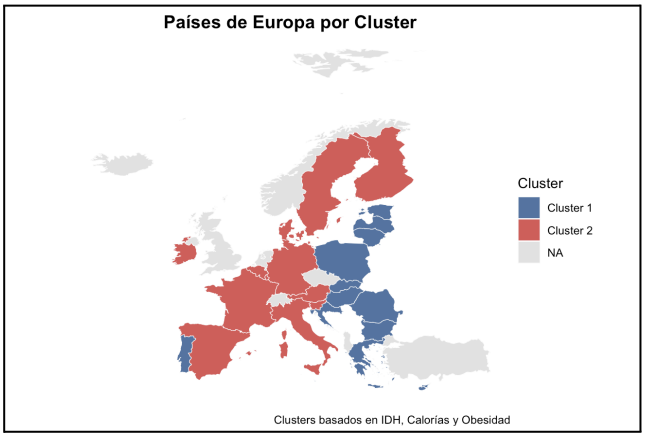


Figure 1: Europe map según Cluster

In the graphic above, countries are colored depending on the cluster they are part of. The first one, with a medium HDI level, is formed by countries that don't ingest too many calories but have a high obesity rate. The other one, on the contrary, is composed of countries with higher HDI values, smaller obesity mean but higher ingestion of calories. Those results are shown in the table in Annex 5.3.8. and have led to investigate more in depth which amount of each type of calories ingest a person for a day in each cluster.

Both clusters consume the same amount of vegetable kilocalories, but the second cluster ingesting animal kilocalories duplicates the amount of the first one as it is shown in the graphics in the Anexo 5.3.9. and 5.3.10. Through Spearman, the method that turned out more suitable, it has been shown that the correlation between Animal.kcal and IDH is about 0.66, which means that countries with higher IDH ingest more Animal.kcal than the rest. On the left side there is a correlation matrix with the variables investigated for each cluster over the years, to analyze its evolution, in graphics in the Anexo 5.3.11., 5.3.12. and 5.3.13. where it seems like IDH depends on taking some calories, especially from animal origin, avoiding the increase of mean obesity . On the contrary, at the right side, there is another correlation matrix with variables related to IDH that are going to be used to make a predictive model of this variable.

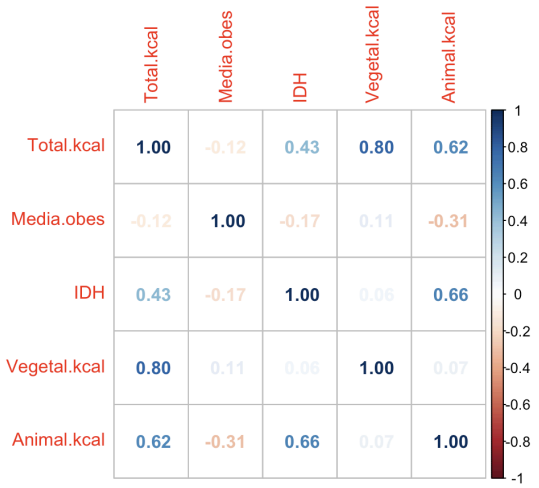


Figure 2: Correlation Matrix

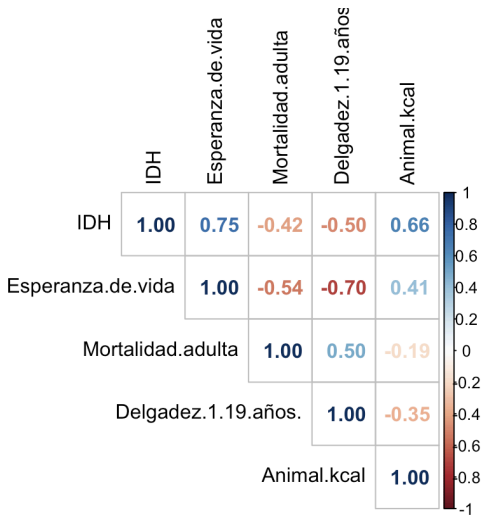


Figure 3: Correlation Matrix

Finally, after classifying in clusters and specifying in a new column which country is part of each cluster, another column is added containing the predicted IDH values. A no linear regression model has been created to figure out if all the variables are enough to predict

an IDH which is in fact based on life expectancy, studies and income (López-Calva, L. 2004). The results in Anexo 5.3.14. are satisfactory because R-squared indicates that 70% of the variability is explained by Life Expectancy, Animal kilocalories and Thinness, after descarting other variables. This model was important in order to check if the IDH classification made sense instead of another one.

IDH = -1,763 + 0,057 Life Expectancy -0.0003 Life Expectancy^2 -0.002 Thinness +0.00015 Animal kcal

3.5. Objective 5

As stated before, the dataset in this analysis is the same as the one used for the objective 2, however unlike above, only año was deleted, as it was not needed to answer the objective. The variables used are: prev.cardiac (prevalence of cardiovascular diseases), media.obesidad (mean obesity), prev.hiper (prevalence of hypertension), prev.diabetes (prevalence of diabetes), porcentaje.gastos_log (the original variable was normalized using a logarithmic transformation due to its severe asymmetry), gasto.total and pais. The aim of this study is to evaluate the relationship between various health indicators and the healthcare expenditure, which will be carried out using a correlation analysis.

The healthcare expenditure is represented in two variables: Porcentaje Gastos and Gasto Total. Porcentaje Gastos is the prevalence of the expenditure on health as a percentage of GDP per capita, whereas Gasto Total is the general government expenditure on health as a percentage of total government expenditure. Both capture different aspects of the healthcare system. By creating a new variable, named 'GastoRelativo', a ratio between both variables is created, allowing a better comparison across the countries and relationship with the illnesses.

In order to choose the appropriate correlation method, it should be taken into account the data distribution of the variables. Considering that variables like GastoRelativo and Prev.cardiac exhibit a skewed distribution (annex 5.3.15) and GastoRelativo and diabetes prevalence have a vast amount of outliers (annex 5.3.16), Spearman seems more adequate than Pearson.

When examining the relationship that exists between health related diseases and healthcare expenditure, the results show weak correlations (annex 5.3.17). The strongest observed correlation is with hypertension, with a negative value of -0.21. This value is not

low enough to consider a relationship linking the variables, so the correlations will be shown separately for each country. With this, it will be possible to examine possible specific patterns.

As shown in figure 4, the relationship involving hypertension prevalence tends to show negative correlations. This indicates that, as the health expenditure increases, the prevalence of hypertension decreases. The contrary is observed with the other three variables, where most countries have positive correlations, meaning that as the health expenditure increases, the prevalence of cardiovascular diseases, diabetes and mean obesity rise accordingly.

The fact that the risk of having these diseases increase can be due to many factors. A plausible explanation is that augmenting the health expenditure could mean an improved capacity to diagnose the three illnesses. This means that there are not necessarily more cases, but an improved medical system.

Several countries such as Jamaica, Cuba and Uruguay show practically no correlation between total health expenditure and the different illnesses. In contrast, countries like Nicaragua and especially Argentina (0.76 for diabetes and 0.75 for obesity) display a clear association between disease prevalence and health expenditure.

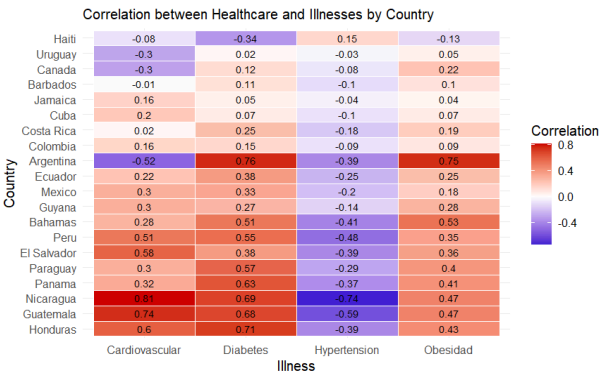


Figure 4: Correlation Matrix by countries

The data is now combined into 3 clusters using Manhattan as distance and median as method for robustness (annex 5.3.18). It is done to observe if the geographical location of countries impacts on the correlation. Countries located near the north are included in the third cluster, where they have practically no correlation between its healthcare expenditure and cardiovascular diseases, diabetes and obesity (annexes 5.3.19 and 5.3.20). Also it can be observed that the countries found in the equator have a significant positive correlation with cardiovascular diseases and diabetes, and a negative correlation, as expected, with hypertension (cluster 2, as seen in annex 5.3.21).

Lastly, studies have shown that as the prevalence of chronic illnesses increases, the health expenditure also increases as a response. This could mean that the observed positive relationship between the sanitary spendings and certain diseases such as diabetes does not necessarily imply a correlation, but rather that as the amount of cases of the illnesses grow, more resources are destined to treat the conditions (Arredondo, 2014).

3.6. Objective 6:

Once the final dataset was obtained, joined by the diabetes, obesity, drugs, alcohol and physical inactivity datasets, an analysis by variable was carried out to determine the distribution of each of the variables studied in order to use the appropriate correlation coefficient (Annex 5.3.24.) . It is also important to mention that two variables were created in order to study a generalized behavior so that the clusters are more coherent and easier to interpret. These are inactivity_index and diabesity_index. The first one is a composite variable that sums standardized values of alcohol consumption, drug use and lack of physical activity. The second, on the other hand, is another standardized composite variable that reflects the prevalence of diabetes and obesity.

On the other hand, in the exploratory analysis it was observed that the drug dataset had atypical but real values that could alter the Pearson correlation index, so this option was discarded and it was decided to use Spearman's correlation coefficient.

With all this said, it was obtained that the most outstanding relationships are: alcohol - inactivity 0.23; obesity - inactivity 0.53; diabetes - inactivity 0.23; diabetes - alcohol 0.19; obesity - drugs -0.33 (negative correlation) (Annex 5.3.25.).

In order to go a little deeper and find out the reality of these relationships and after scaling the data it was decided to do clustering with 4 clusters (k-means) , decided by the elbow method.(Annex 5.3.26.). Although the clusters give us information on how the data are grouped and distributed, in order to have a better visualization and to be able to project in 2 dimensions, PCA has also been used.(Annex 5.3.27.), (Annex 5.3.28.).

As a result, the 4 clusters consist of: (Annex 5.3.29.)

Cluster 1: countries with high physical activity, low alcohol consumption, low obesity and diabetes, but medium-high levels of drug use. These are Zambia, Guatemala, Burkina Faso, etc. This group seems to be made with countries with an active lifestyle, however health conditions are uncontrolled. Drug use is the main issue in this cluster.

Cluster 2: countries like Samoa or Tonga have a very low level in most variables. The outliers in the obesity variable stand out because they are the countries with the highest obesity rates in the world. Obesity is the dominant problem in this cluster.

Cluster 3: countries with moderate alcohol and drug consumption, moderate levels of obesity and inactivity, and low levels of diabetes. These countries are Andorra, Cuba, Costa Rica, El Salvador, etc. and they seem to have a more balanced profile, with more controlled habits. This could be because they are more developed countries than those in the previous clusters, with stronger public health systems or cultural habits.

Cluster 4: countries with high levels of obesity, diabetes, alcohol and drugs, such as Argentina, Colombia, Chile or Serbia. This cluster is overall unhealthy, possibly due to typical habits from living in big cities, their diets and higher exposure to processed foods and substances.

3.7. Reflection on the privacy and legality

For the study of each objective, the various datasets obtained contain publicly accessible and open data. Considering that these datasets come from official entities such as the WHO, it is deemed that the privacy was not compromised at any point.

Moreover, the protection of individual privacy has limited the scope of this project, as it has only been possible to conduct an analysis at country levels using aggregated data, rather than with population subgroups or individual citizens. Another important limitation previously mentioned was the limited compatibility of countries across the distinct datasets.

4. Conclusions

After all the analyses carried out using maps, charts, clustering, correlations and many other methods, it was possible to draw several conclusions regarding alimentation, country development, their illnesses and spendings. Various aspects on the topic were studied, some on American countries and others on European countries.

Regarding the influence of the level of economic development on dietary habits we conclude that the study reveals that developed countries consume more animal fats and plant-based calories; developing countries consume more animal-based calories and sugars. The temporal analysis shows an increase in animal calories in both groups, higher in developing countries. PCA and hierarchical clustering verify the economic clustering of dietary habits. Linear regression finally shows that HDI and GDP per capita are important predictors of dietary variety and composition.

With respect to objective 2, after the analyses, the final conclusion is that there is a positive correlation between diabetes and obesity. That is, those countries with high obesity tend to have high diabetes. The creation of the diabesity_index has made it possible to represent the situation of each country in relation to both diseases, with some countries in the Americas and Oceania standing out in particular. Taken together, these results affirm that obesity is a key factor in the prevention and detection of diabetes.

Following the analysis of the relationship between dietary patterns, cardiovascular health and life expectancy during the period 2010-2014, three groupings/profiles of dietary habits and health indicators have been distinguished, where it can be seen which countries belong to each. In this way, when comparing these profiles with life expectancy, the need to extend this study with other variables that can help to deepen the results has been observed. Nevertheless, certain behaviours can be observed that show that a balanced, high-calorie diet (especially in Frutas, Verduras and Grasas Animales) and a prevalence of cardiovascular disease is associated with a better life expectancy. Whereas dietary and health instability over 2010-2014 show lower levels of longevity. In addition, a general stability has been observed among the countries belonging to a cluster, which reinforces the consistency of the profiles of each cluster. Finally, this analysis opens the door to a possible relationship between fat consumption and a country's political or economic situation.

Taking into account how the ingest of kilocalories and obesity affects the HDI, it can be stated that even though countries that consume more kilocalories tend to have a higher HDI value, there are different types of kilocalories and it's important to balance its quantities depending on it. The proportion of vegetable kilocalories in front of animal ones has to be so much greater, but that does not affect the fact that consuming animal kilocalories improves the HDI. Moreover, less thinness and more life expectancy also improve the HDI, which makes it even more complete for choosing suitable countries to live in. Countries in Europe like Spain, France, Italy, Germany, Sweden and Finland that have turned out to have higher IDH.

As for the relationship between healthcare expenditure and several nutritional based diseases, such as hypertension, diabetes, cardiovascular illnesses and obesity varies across the different American countries. In general, there is a negative correlation with hypertension prevalence, which suggests that increasing the spending can help reduce its prevalence. However, as for the other three diseases, they tend to show positive correlations, which may happen because, as shown in several studies, a higher health spending leads to improved detection and treatment of the disease as their number of cases rise, rather than having a causal relationship. The clustering analysis indicates that the country's location can affect the correlations, where northern countries tend to show no correlation between illnesses and health expenditure, while equatorial countries manifest positive correlation with diabetes, cardiovascular diseases and obesity, while being negative with hypertension.

After the analysis of objective 6, it was found that alcohol consumption and lack of physical activity have a positive relationship with diabetes and obesity; on the other hand, drug consumption shows a weak or inverse relationship in the case of obesity. The PCA has been useful to show that obesity and physical inactivity are highly correlated, and clustering has shown that countries with a high rate of obesity tend to be the least active and tend to have the lowest alcohol consumption. In summary, the data affirm that bad habits, especially sedentary lifestyles, significantly influence the levels of obesity and diabetes.

4.1. Lessons learned for future projects

After the elaboration of this project, all members of the group have acquired new knowledge about varied aspects of exploratory analysis and the studied subject. Amongst

what was learned, it can be highlighted the new data search skills, cleaning and data transformation. R Studio technical aptitudes have also been gained, like the discovery of new libraries and methods of visual representation, for example ggplot2.

Furthermore, due to the delivery deadlines and the number of team members, our organizational capacity has improved. This has been one of the most important lessons learned, as it will be crucial for future occasions to successfully carry out new projects.

In addition, the subject of the project studied has offered all participants a deeper understanding on the socio-economical, alimentary and sanitary contexts of the distinct studied countries. All of this was achieved through a thorough research across different information sources, in order to improve the overall level of comprehension.

4.2. Future research proposal

After the completion of this project, the results are considered highly relevant due to their usefulness to society. The topics addressed have a great impact on today's world. Nevertheless, it is considered appropriate to search for new aspects that may influence the relationship between dietary habits and various diseases. For example, it is considered useful for future research into areas such as access to healthcare or educational level, as these factors can greatly influence the overall health of the population.

5. Annexes

5.1. Hito 1

Enlace donde se encuentra el Hito I: Data sources:

https://docs.google.com/document/d/1tIXs6D0mADc6FthKDB6LbSYuKeFXnRNx/edit?usp=sharing&ouid=113522848902946787022&rtpof=true&sd=true

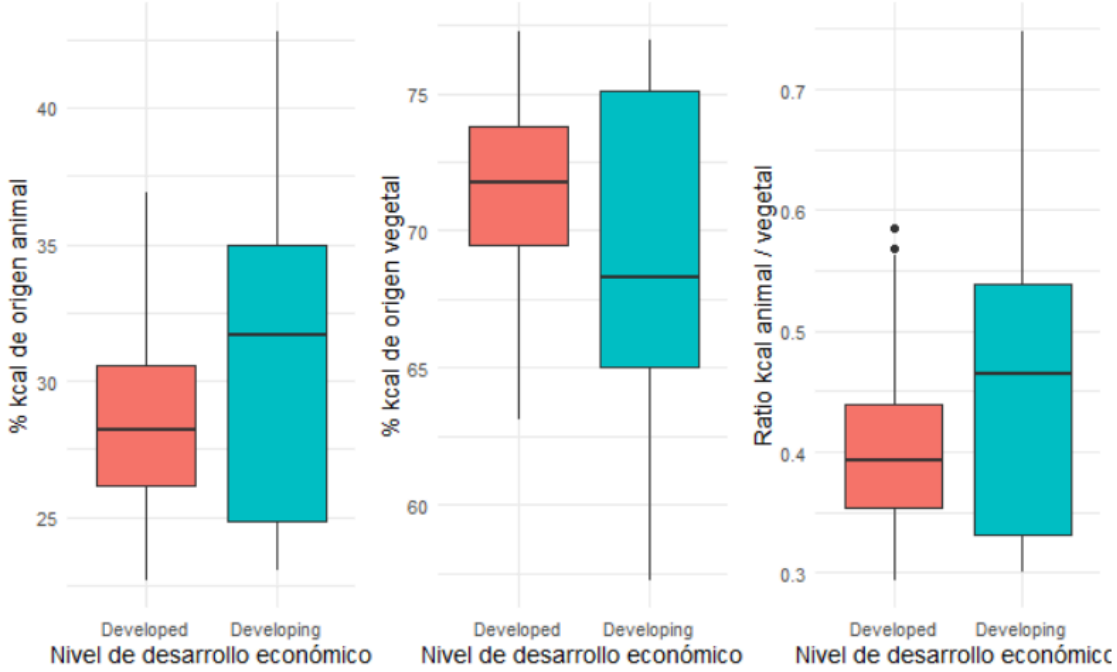
5.2. Hito 2

Enlace donde se encuentra el Hito II: Data integration and transformation:

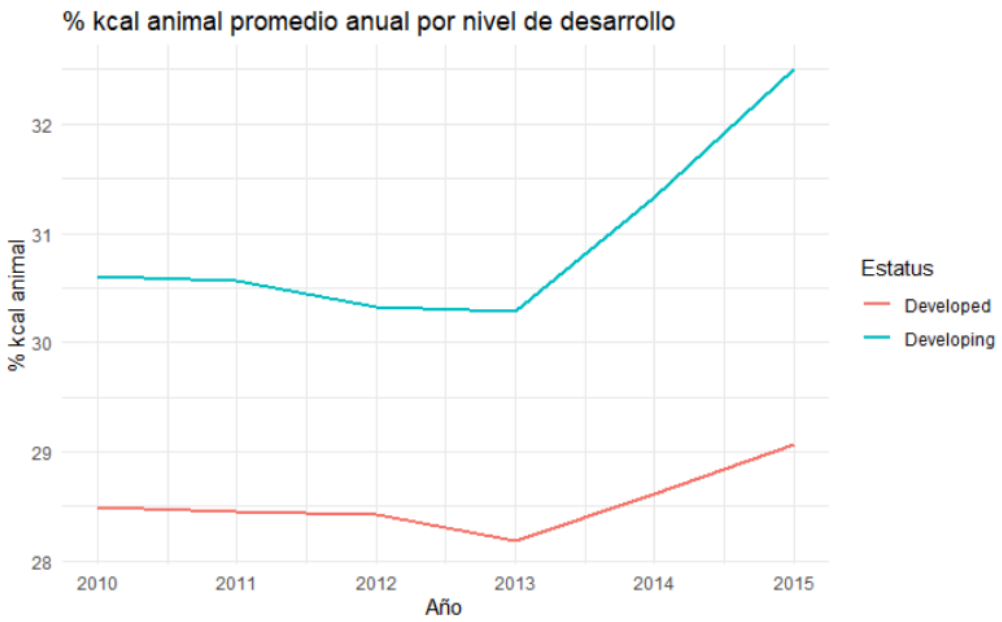
https://docs.google.com/document/d/1o9-e0HhxwbVDT-RFFJjbZUQqkQTa7Hvs/edit?usp=sharing&ouid=113522848902946787022&rtpof=true&sd=true

5.3. Graphics annexes

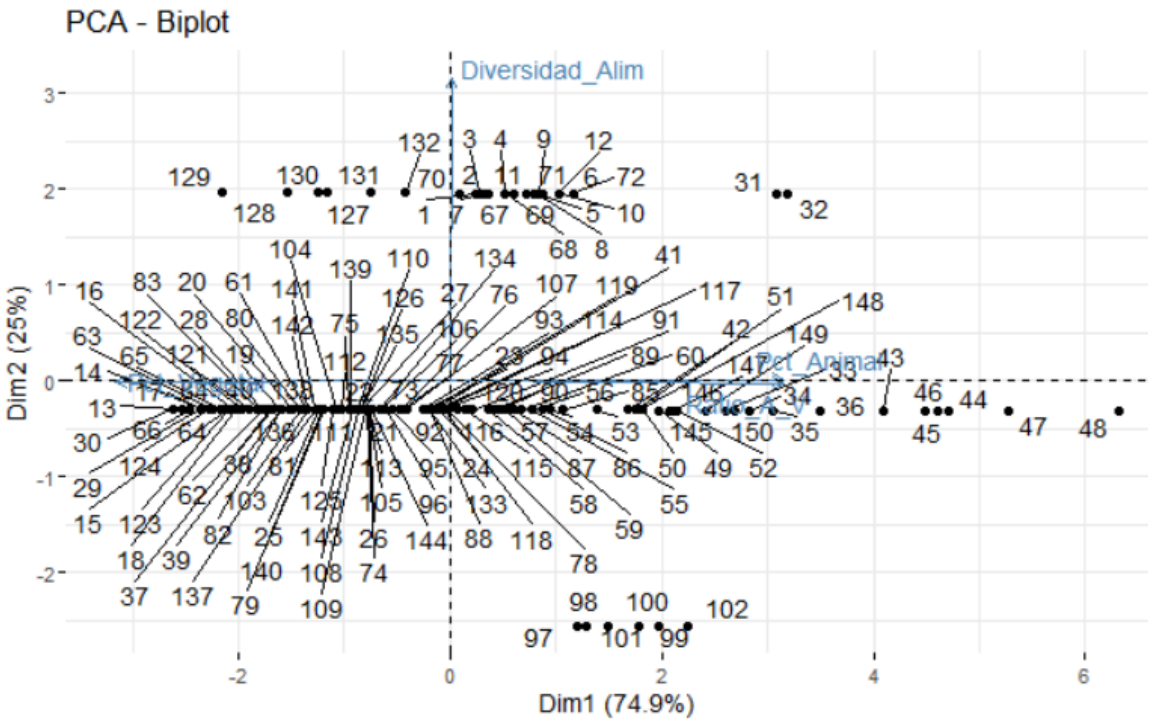
5.3.1. Comparison graphs developed and developing

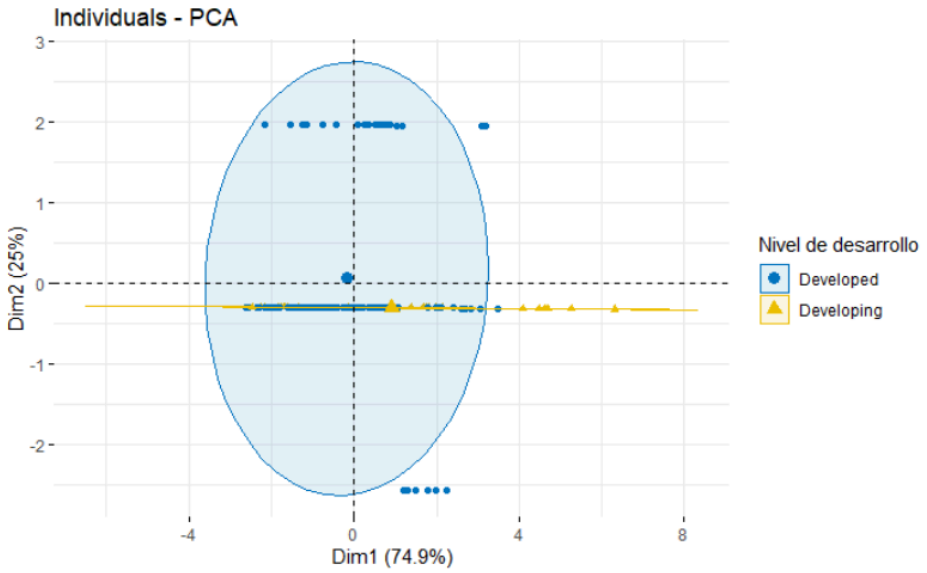


5.3.2. Table %kcal animal annual average

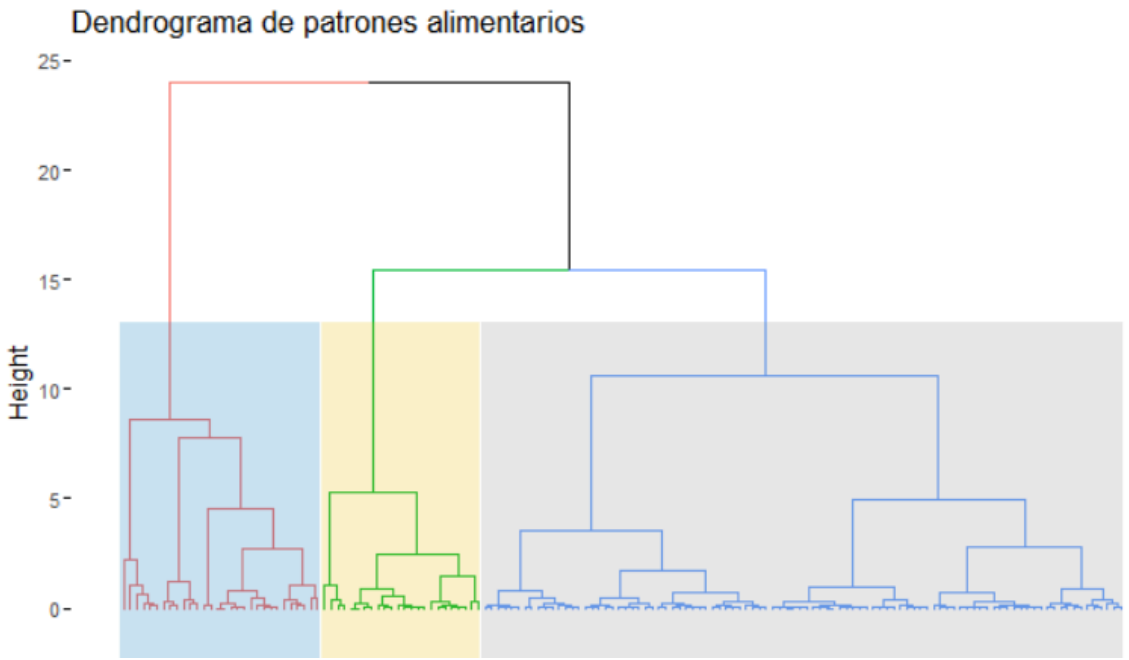


5.3.3. PCA





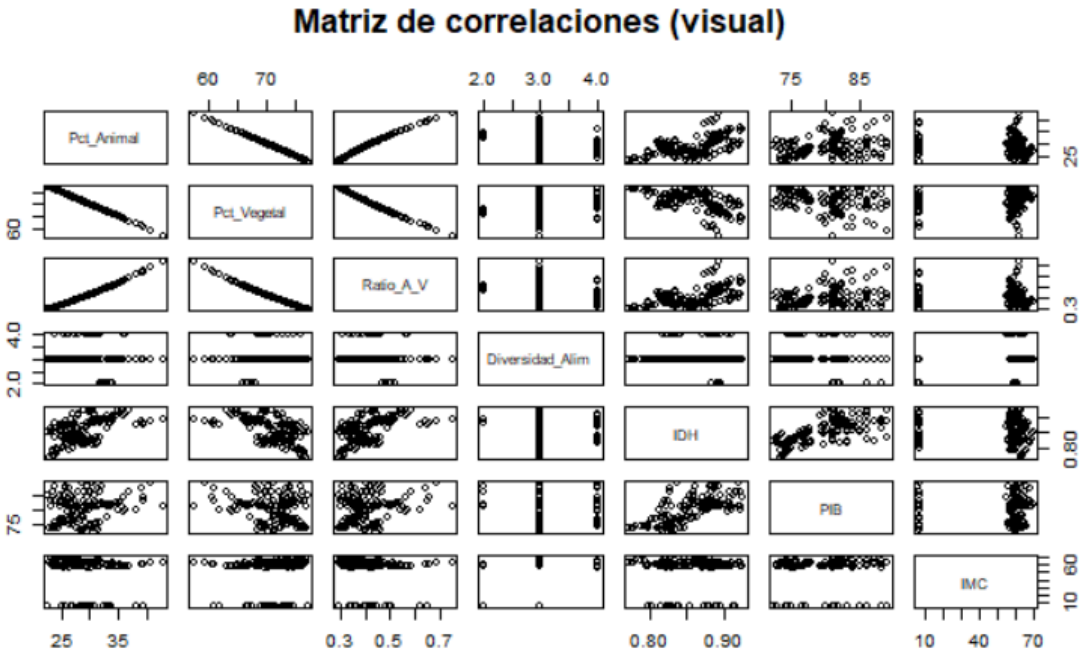
5.3.4. Dendrogram of food patterns



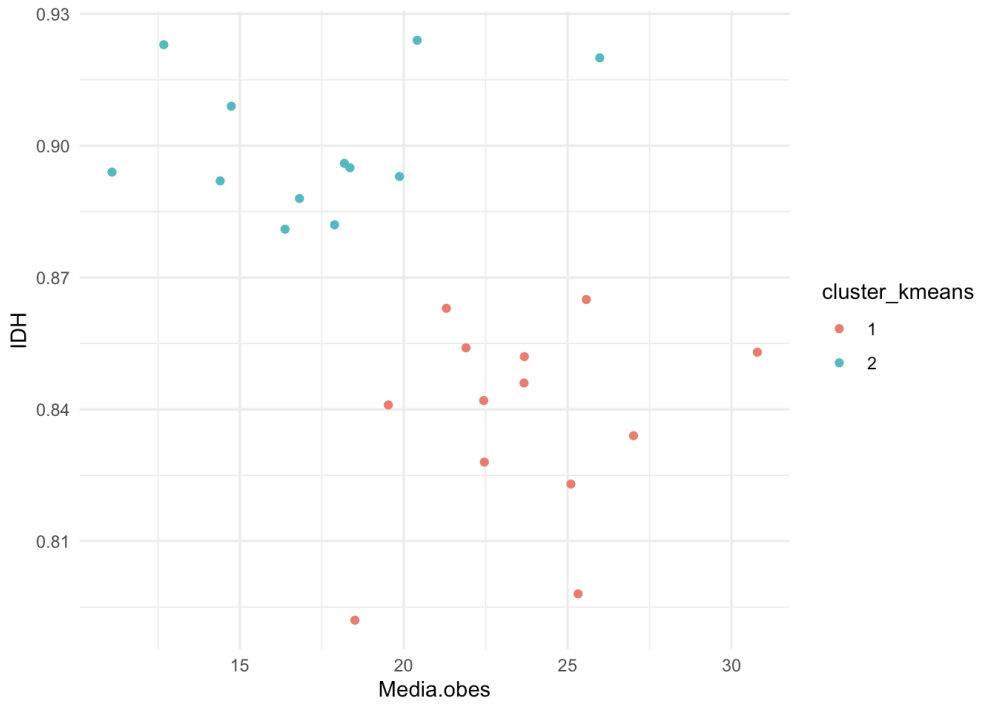
5.3.5. Linear regression model

```
Call:
lm(formula = Pct_Animal ~ IDH + PIB + IMC + Año, data = obj1_clean)

Residuals:
   Min     1Q Median     3Q    Max 
-6.062 -2.529 -0.091  2.056 11.885 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept)  71.79256  322.08123   0.223   0.8239    
IDH          73.88916    9.94247   7.432 8.53e-12 ***
PIB          -0.20806    0.08563  -2.430   0.0163 *  
IMC          -0.02702    0.01386  -1.949   0.0532 .  
Año          -0.04382    0.16053  -0.273   0.7853    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 3.302 on 145 degrees of freedom
Multiple R-squared:  0.3387,	Adjusted R-squared:  0.3204 
F-statistic: 18.56 on 4 and 145 DF,  p-value: 2.444e-12
```

5.3.6. Correlation matrix



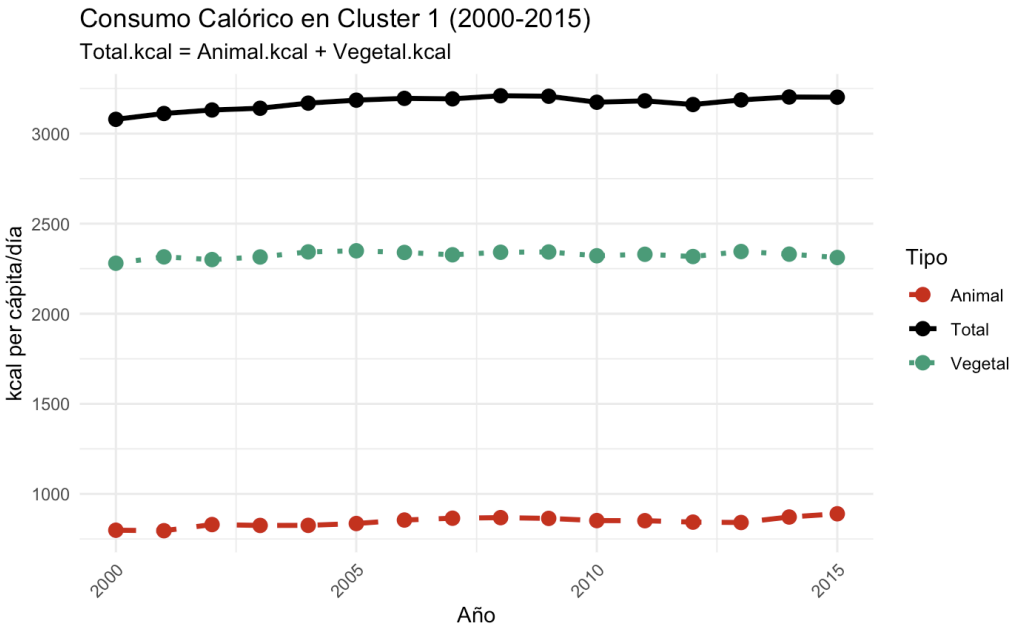
5.3.7. Scatter plot of HDI clusters



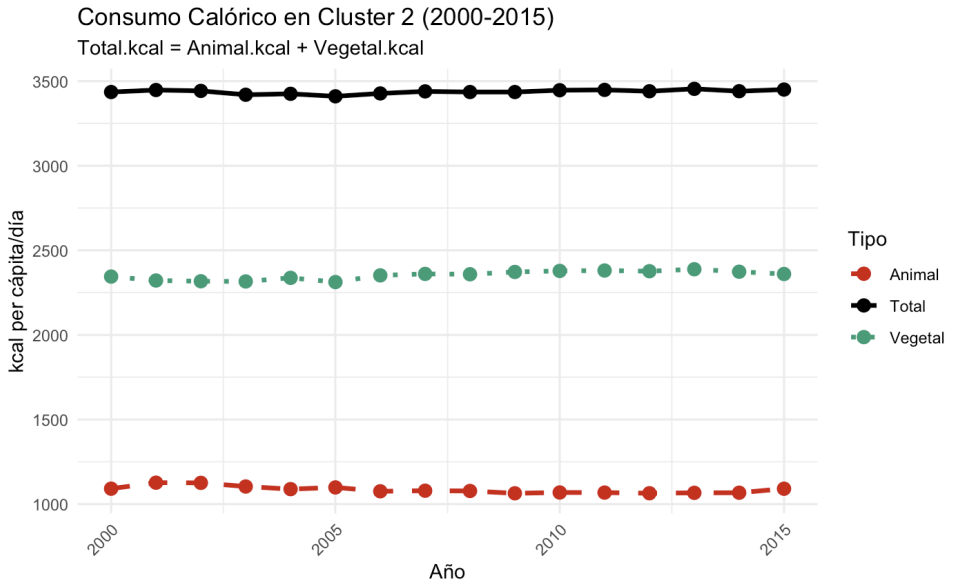
5.3.8. Coordinate table

```
          IDH Total.kcal Media.obes
1 -0.8053475 -0.4546968  0.6412679
2  0.8724598  0.4925882 -0.6947069
```

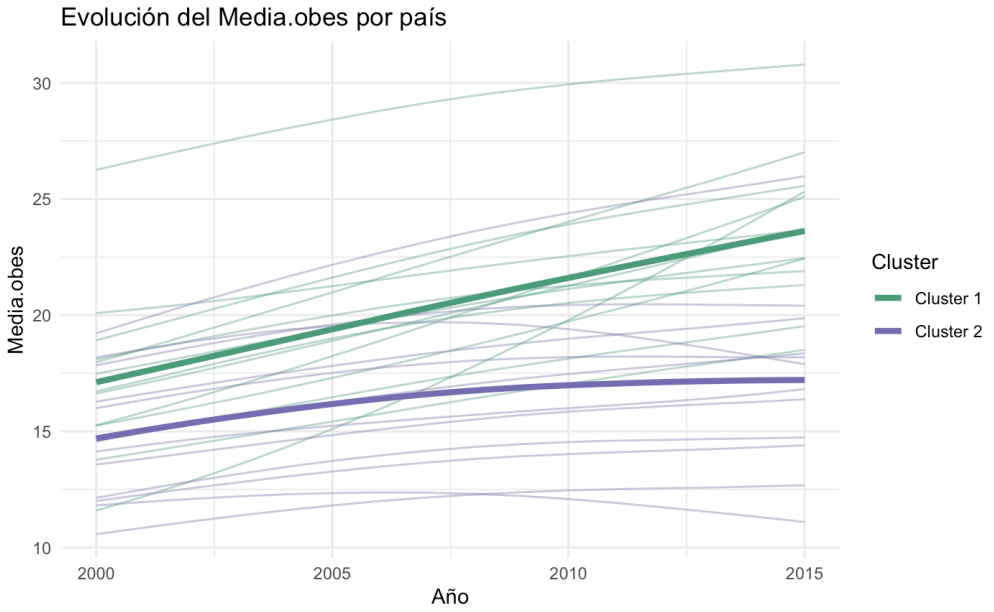
5.3.9. Line graph of Cluster 1 Kcal



5.3.10. Line graph of Cluster 2 Kcal

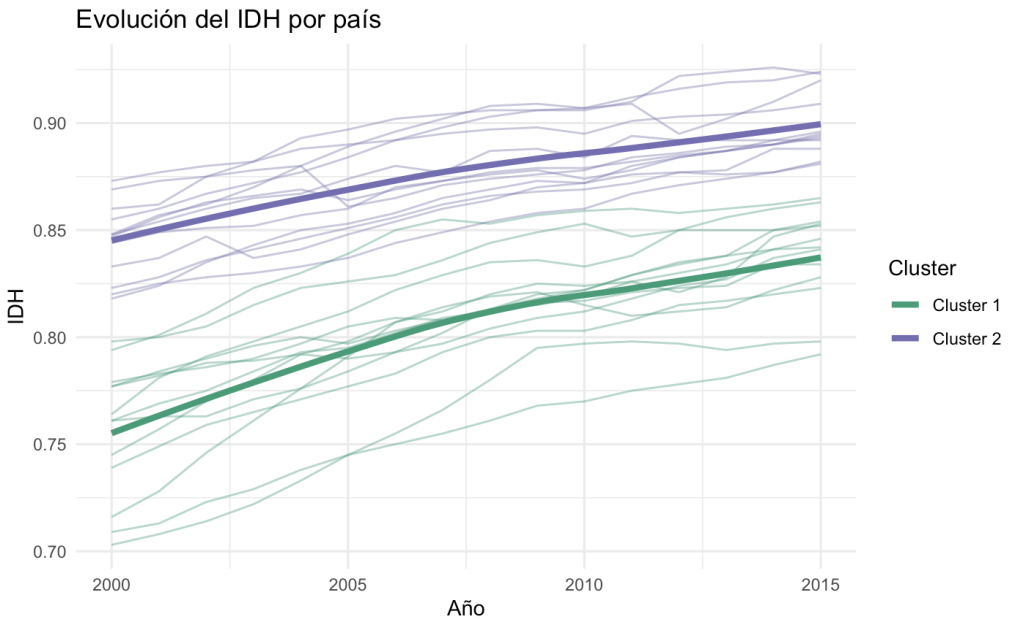


5.3.11. Line graph of Media.obes (2000-2015)

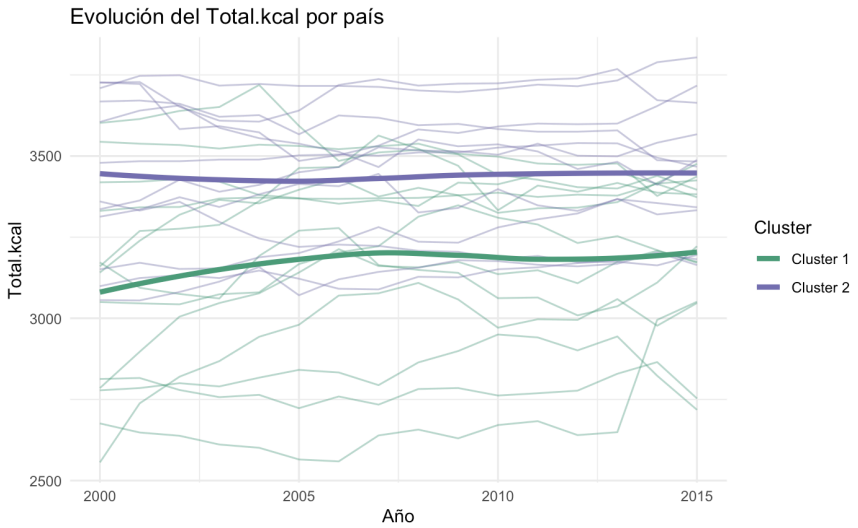


Se reali

5.3.12. Line graph of IDH (2000-2015)



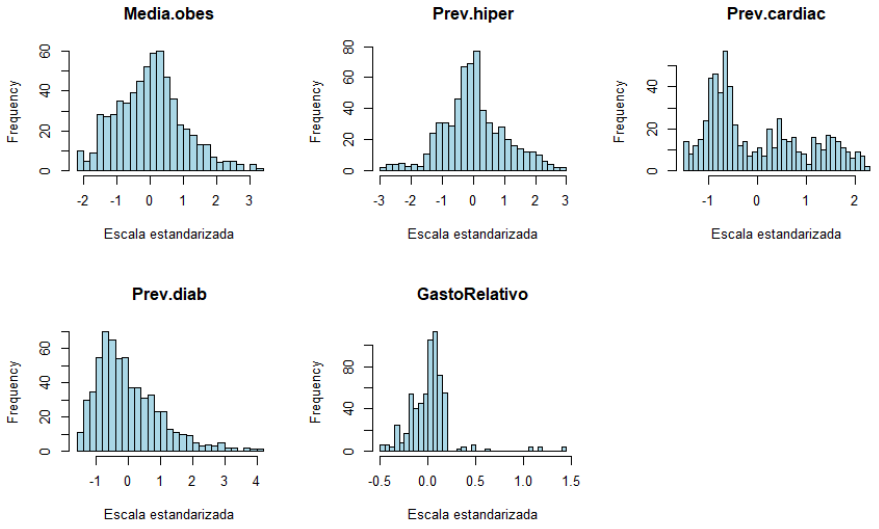
5.3.13. Line graph of Total.kcal (2000-2015)



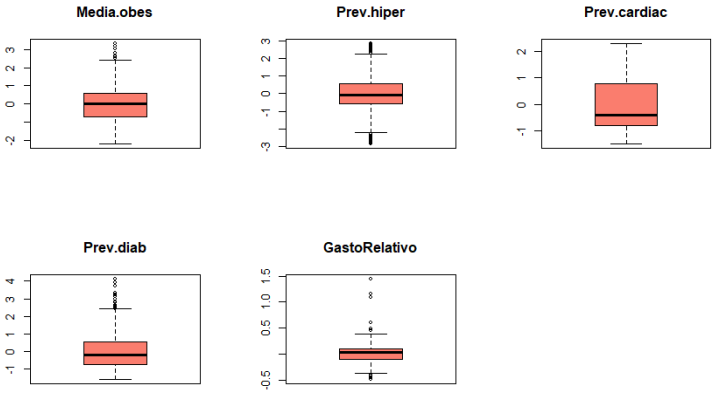
5.3.14. Regression model summary

```
Call:
lm(formula = IDH ~ Esperanza.de.vida + Delgadez.1.19.años. +
    I(Esperanza.de.vida^2) + Animal.kcal, data = train)

Residuals:
      Min        1Q    Median        3Q       Max
-0.098758 -0.015570  0.003588  0.019793  0.051760

Coefficients:
                        Estimate Std. Error t value Pr(>|t|)
(Intercept)           -1.763e+00  4.938e-01  -3.571  0.00041 ***
Esperanza.de.vida      5.740e-02  1.225e-02   4.686 4.13e-06 ***
Delgadez.1.19.años.   -2.800e-03  3.019e-03  -0.927  0.35444
I(Esperanza.de.vida^2) -3.241e-04  7.592e-05  -4.270 2.58e-05 ***
Animal.kcal            1.058e-04  9.479e-06  11.167  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.02664 on 320 degrees of freedom
Multiple R-squared:  0.7005,    Adjusted R-squared:  0.6967
F-statistic: 187.1 on 4 and 320 DF,  p-value: < 2.2e-16
```

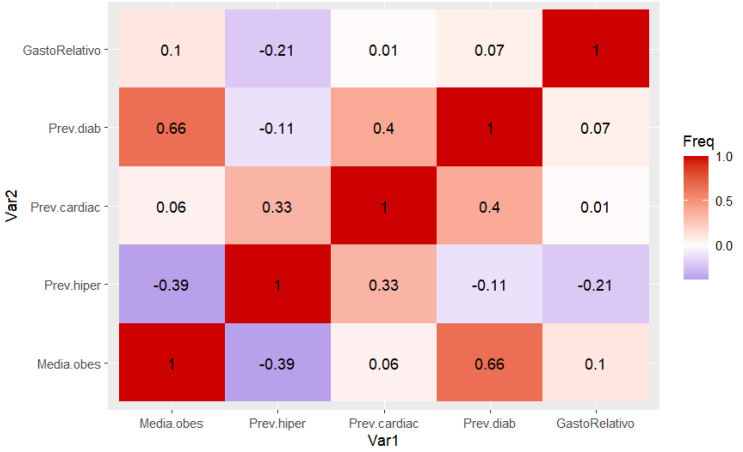
5.3.15. Distribution of variables from objective 5



5.3.16. Boxplots of variables from objective 5



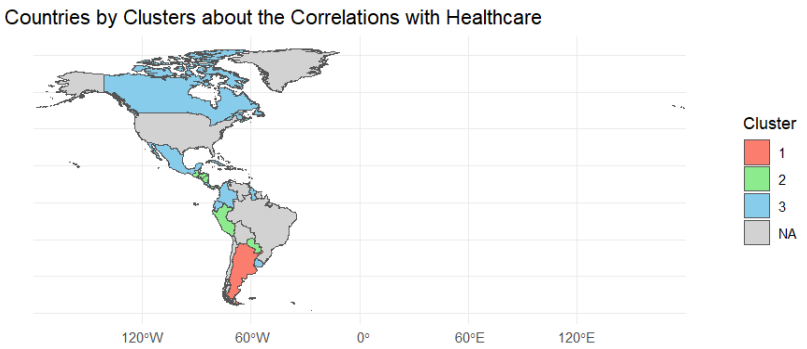
5.3.17. Correlation between illnesses and health expenditure



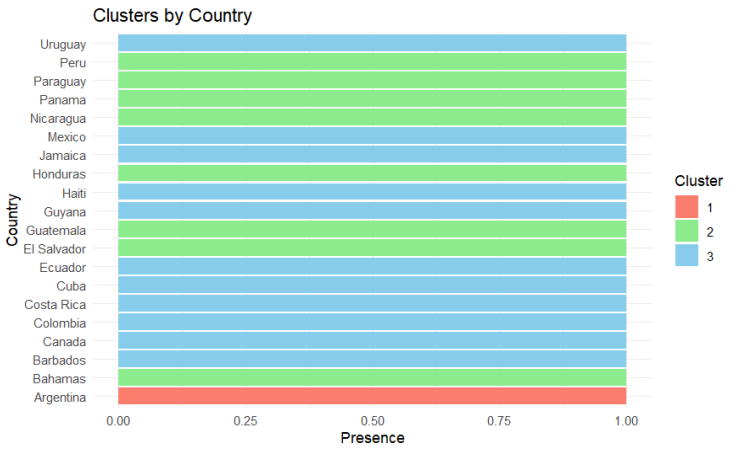
5.3.18. Number of clusters chosen in objective 5

```
                    the measure.

*******************************************************************
* Among all indices:
* 6 proposed 2 as the best number of clusters
* 11 proposed 3 as the best number of clusters
* 1 proposed 4 as the best number of clusters
* 1 proposed 5 as the best number of clusters
* 4 proposed 6 as the best number of clusters

                   ***** Conclusion *****

* According to the majority rule, the best number of clusters is  3


*******************************************************************
```

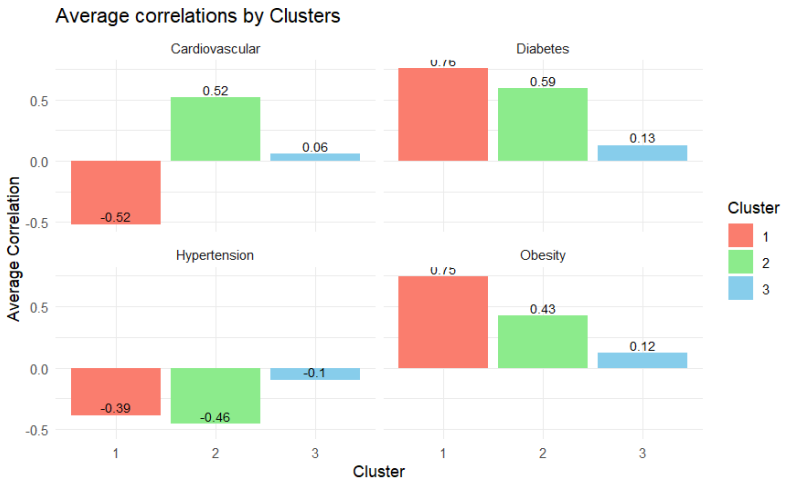
5.3.19. Map of countries by cluster



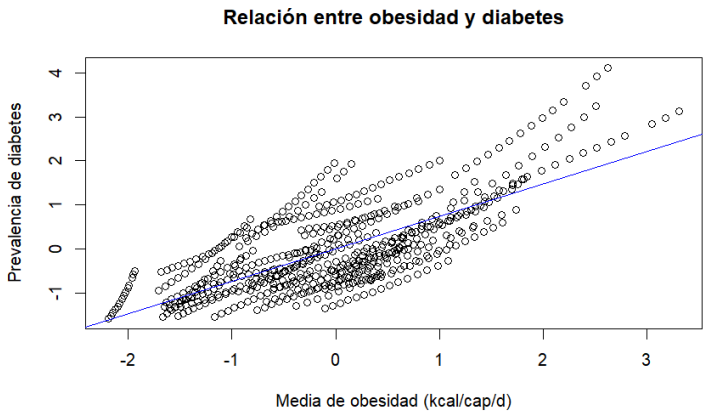
5.3.20. Graph of countries by cluster



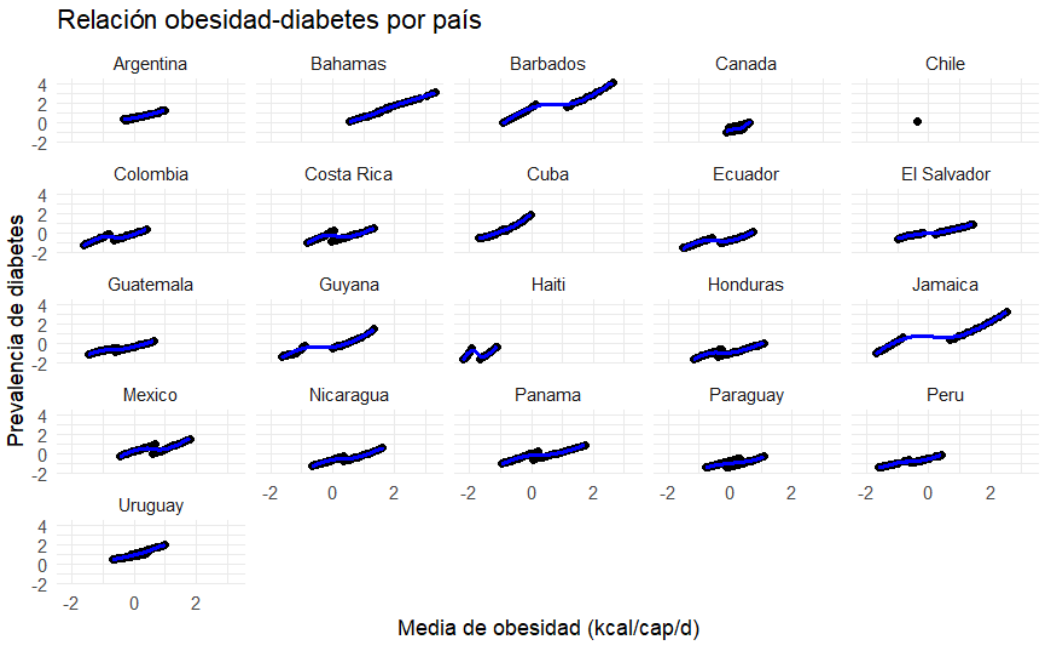
5.3.21. Correlations by clusters



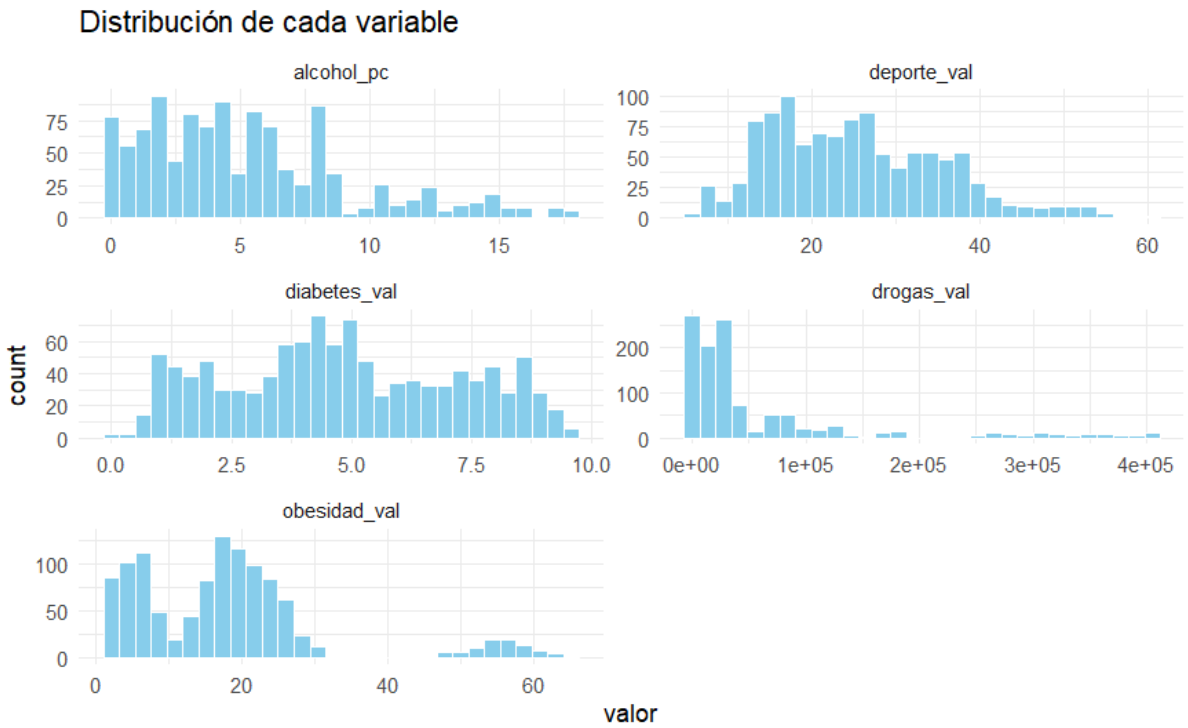
5.3.22. Scatterplot of diabetes versus obesity



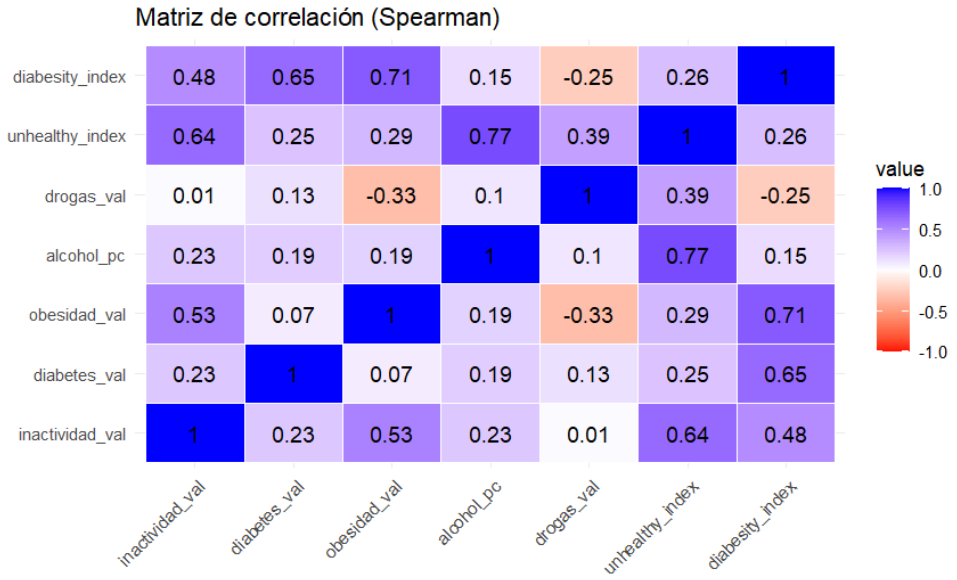
5.3.23. Obesity vs. Diabetes: Scatterplots by Country



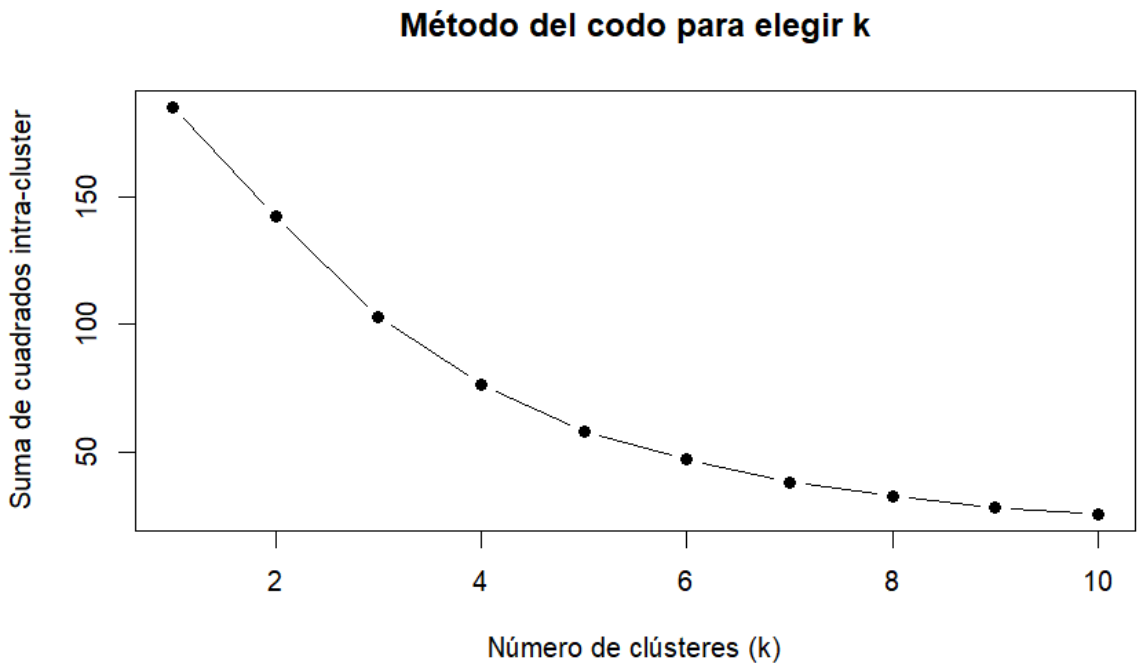
5.3.24. Distribution of variables from objective 6



5.3.25. Correlation Matrix Spearman's



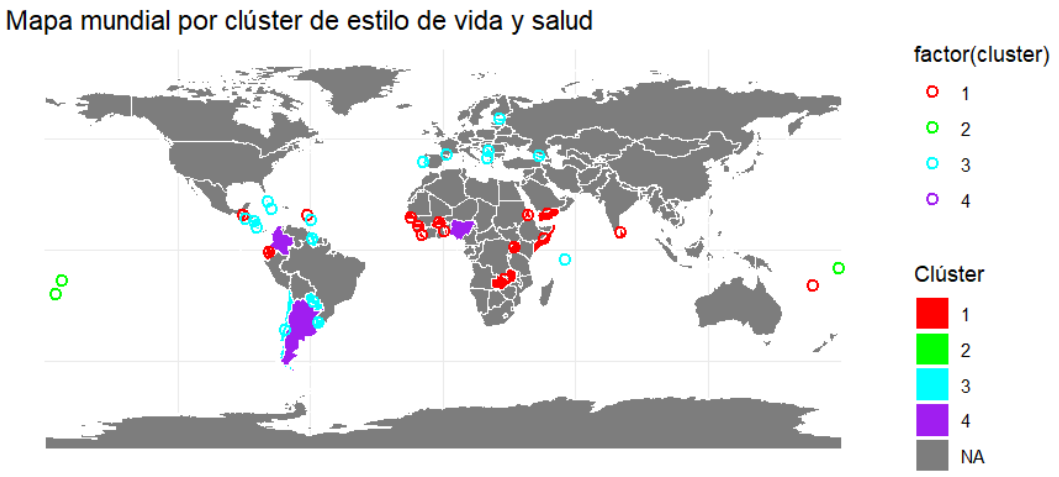
5.3.26. Elbow method to decide clusters in objective 6



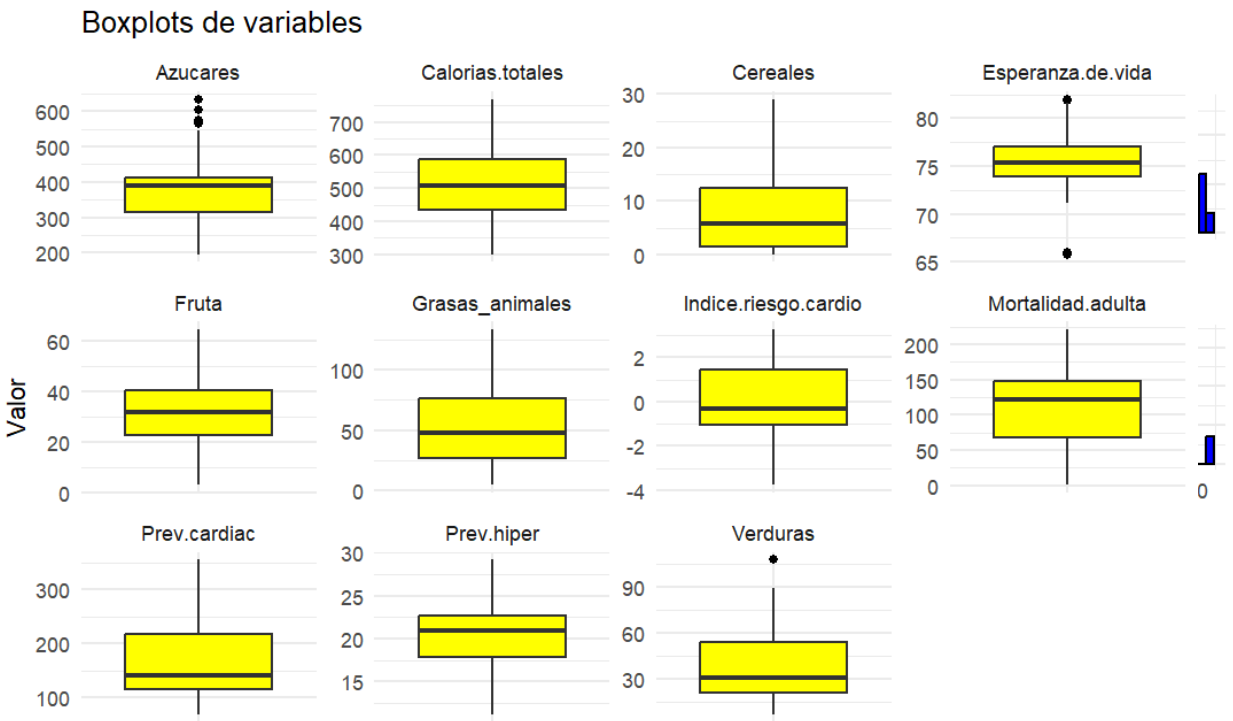
5.3.27. Clustering Visualization on PCA Space

5.3.28. Clustering visualization on PCA space with scores (biplot)

5.3.29. World map with clusters by countries (objective 6)

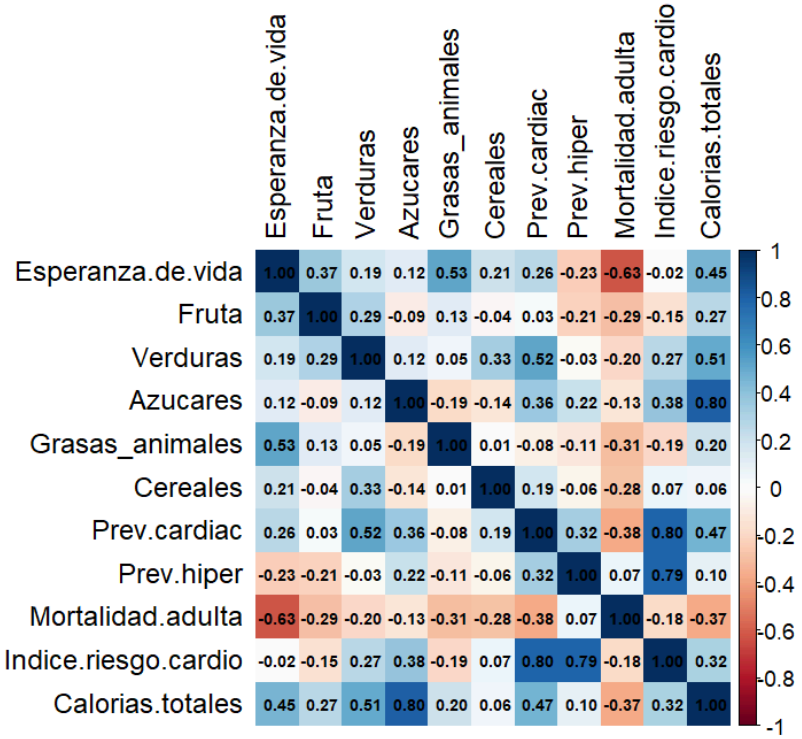


5.3.30. Distribution of variables from objective 3



5.3.31. Boxplots of variables from objective 3

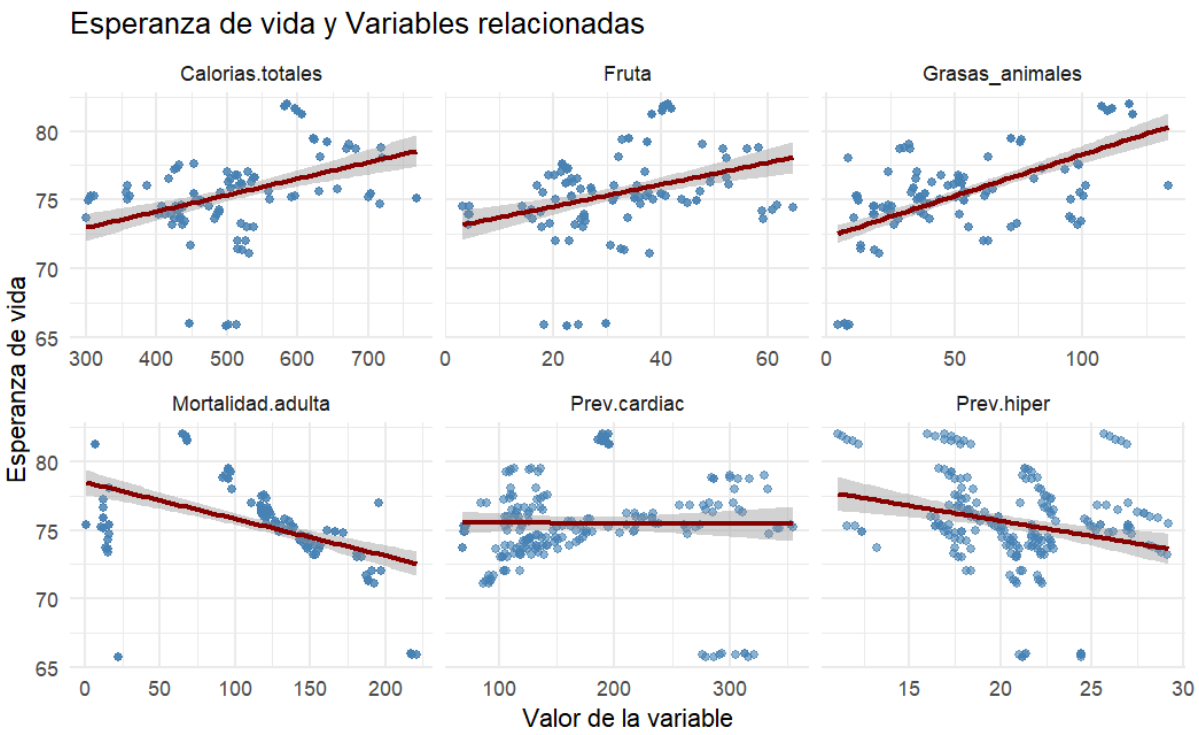
5.3.32. Correlation Matrix from objective 3



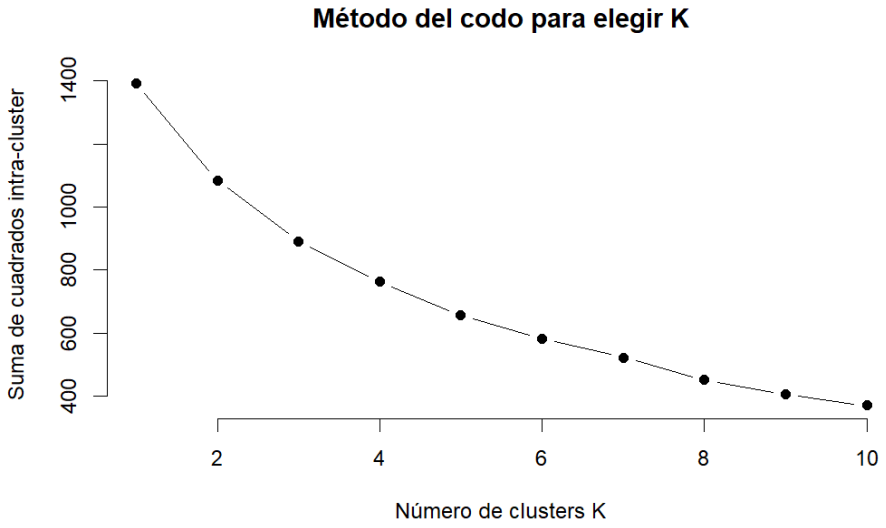
5.3.33. Regression model summary from objective 3

```
Call:
lm(formula = Esperanza.de.vida ~ Fruta + Verduras + Azucares +
    Grasas_animales + Cereales + Prev.hiper + Prev.cardiac +
    Mortalidad.adulta + Indice.riesgo.cardio + Calorias.totales +
    factor(Año), data = objetivo)

Residuals:
    Min      1Q  Median      3Q     Max
-6.4865 -1.5521  0.3312  1.7064  3.6317

Coefficients: (2 not defined because of singularities)
                       Estimate Std. Error t value Pr(>|t|)
(Intercept)          70.7137815  1.4474805  48.853  < 2e-16 ***
Fruta                 0.0585262  0.0121864   4.803 3.20e-06 ***
Verduras             -0.0304777  0.0097696  -3.120 0.002098 **
Azucares              0.0109821  0.0018894   5.812 2.61e-08 ***
Grasas_animales       0.0583301  0.0050149  11.631  < 2e-16 ***
Cereales              0.1081662  0.0252648   4.281 2.97e-05 ***
Prev.hiper           -0.1617912  0.0435486  -3.715 0.000268 ***
Prev.cardiac         -0.0004612  0.0030897  -0.149 0.881496
Mortalidad.adulta    -0.0081779  0.0032167  -2.542 0.011823 *
Indice.riesgo.cardio         NA         NA      NA       NA
Calorias.totales             NA         NA      NA       NA
factor(Año)2011       0.0532524  0.5314412   0.100 0.920290
factor(Año)2012      -0.0112911  0.4653892  -0.024 0.980670
factor(Año)2013       0.3537927  0.4816204   0.735 0.463510
factor(Año)2014       0.7883178  0.4724478   1.669 0.096874 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 2.13 on 187 degrees of freedom
Multiple R-squared:  0.631,     Adjusted R-squared:  0.6074
F-statistic: 26.65 on 12 and 187 DF,  p-value: < 2.2e-16
```

5.3.34. Distribution of Esperanza de Vida by variable



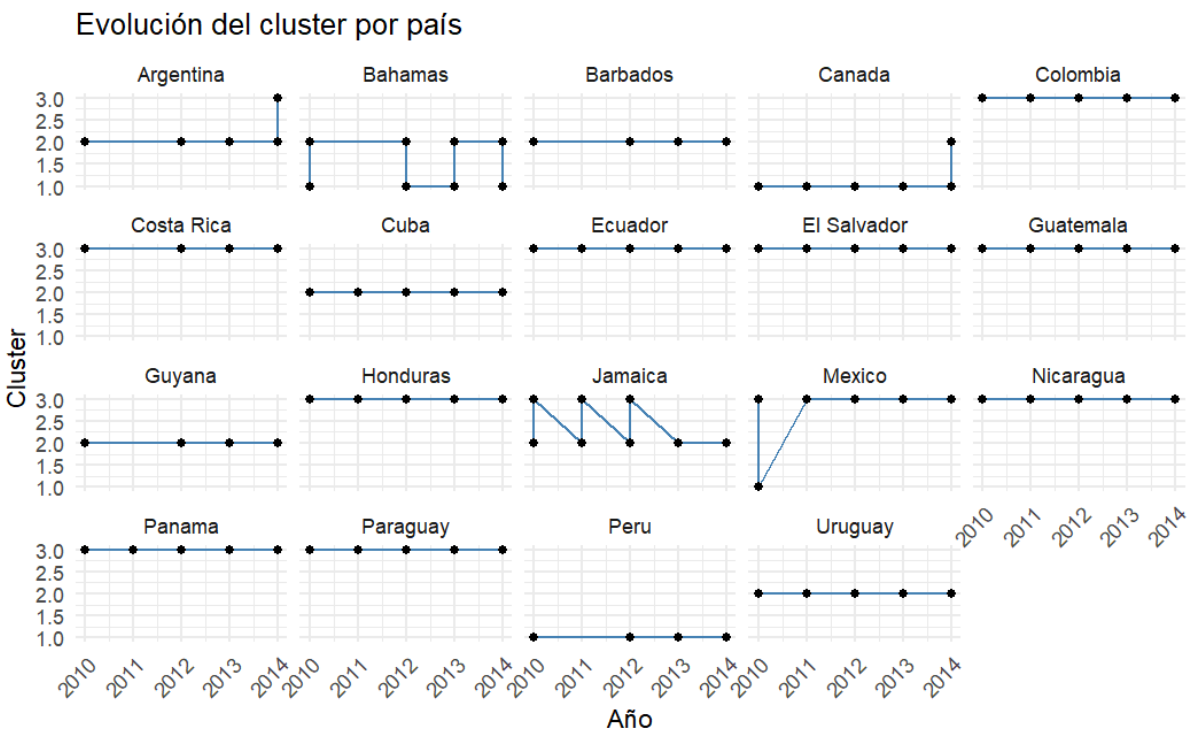
5.3.35. Elbow method from objective 3



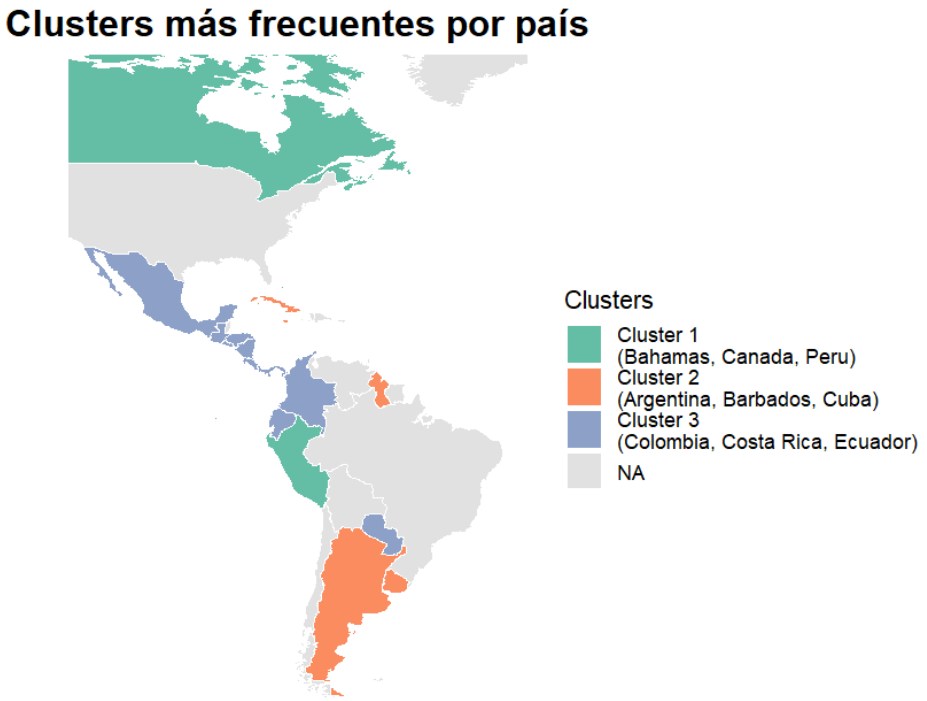
5.3.36. NbClust from objective 3

```
*** : The Hubert index is a graphical method of determining the number of clusters.
                In the plot of Hubert index, we seek a significant knee that
corresponds to a
                significant increase of the value of the measure i.e the significant
peak in Hubert
                index second differences plot.

*** : The D index is a graphical method of determining the number of clusters.
                In the plot of D index, we seek a significant knee (the significant
peak in Dindex
                second differences plot) that corresponds to a significant increase
of the value of
                the measure.

*******************************************************************
* Among all indices:
* 6 proposed 3 as the best number of clusters
* 3 proposed 4 as the best number of clusters
* 2 proposed 5 as the best number of clusters
* 6 proposed 6 as the best number of clusters
* 6 proposed 7 as the best number of clusters

                   ***** Conclusion *****

* According to the majority rule, the best number of clusters is  3
```

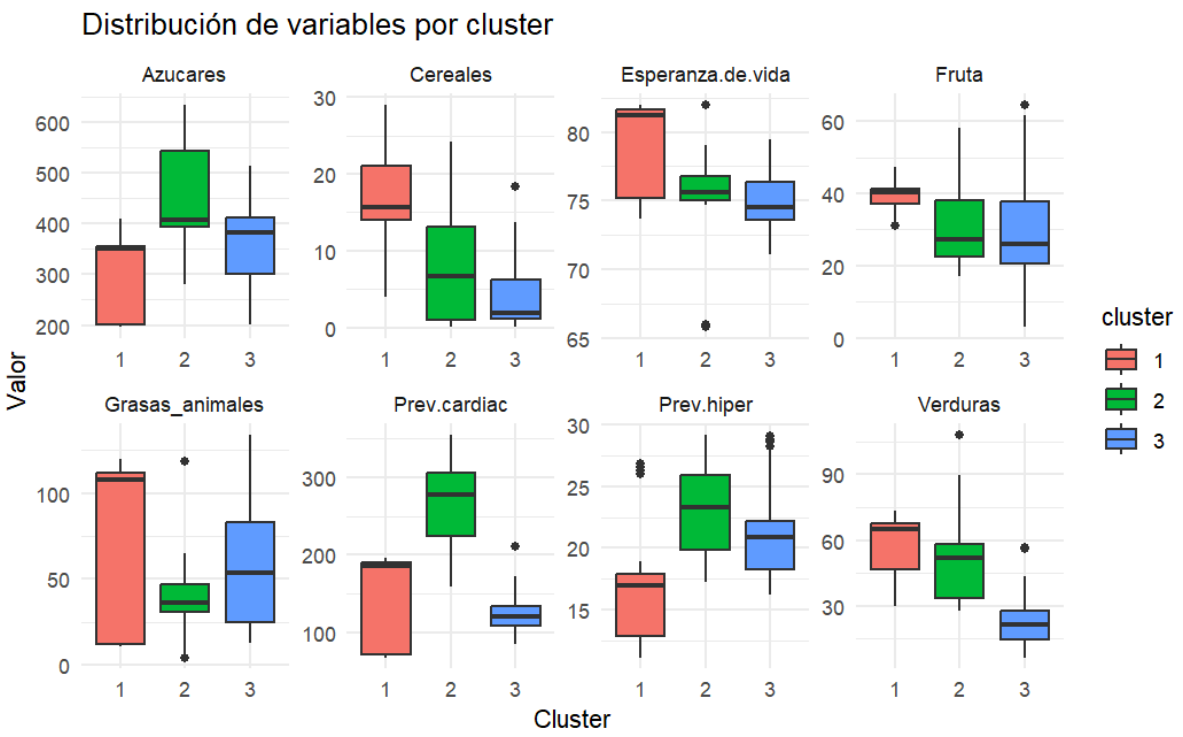
5.3.37. Evolution of countries by cluster



5.3.38. Map with most common countries per cluster



5.3.39. Distribution of variables by cluster

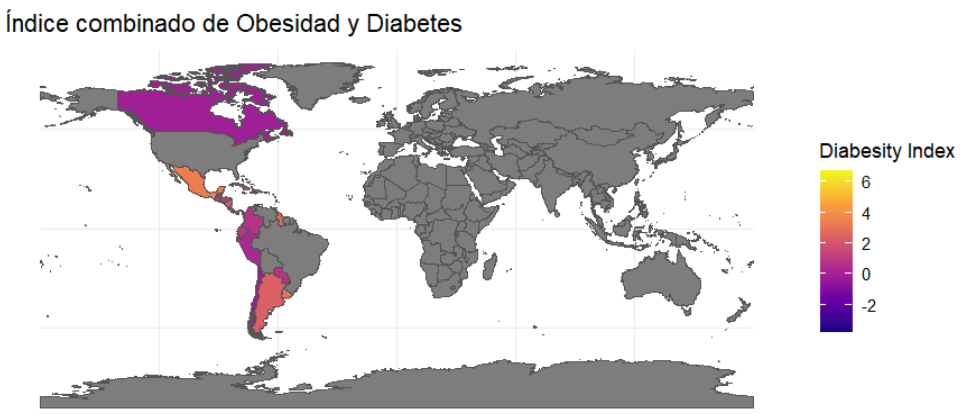


5.3.40. Distribution of gender by cluster

5.3.41. Variables of objective 3

```
'data.frame':   200 obs. of  12 variables:
 $ Pais             : chr  "Argentina" "Argentina" "Argentina" "Argentina" ...
 $ Año              : int  2010 2010 2012 2012 2012 2013 2013 2014 2014 2014 ...
 $ Esperanza.de.vida: num  75.5 75.5 75.9 75.9 75.9 76 76 76.2 76.2 76.2 ...
 $ Fruta            : num  18 18 27.1 27.1 27.1 ...
 $ Verduras         : num  31.4 31.4 31.9 31.9 31.9 ...
 $ Azucares         : num  403 403 405 405 405 ...
 $ Grasas_animales  : num  33.1 33.1 34 34 34 ...
 $ Cereales         : num  4.86 4.86 6.78 6.78 6.78 ...
 $ Prev.cardiac     : num  238 235 232 223 232 ...
 $ Prev.hiper       : num  29.2 19.5 18.7 18.7 28.6 28.2 18.4 27.9 18 27.9 ...
 $ Mortalidad.adulta: int  121 121 12 12 12 119 119 118 118 118 ...
 $ Sexo             : chr  "Men" "Women" "Women" "Women" ...
             Pais                Año Esperanza.de.vida             Fruta          Verduras
                0                  0                 0                 0                 0
         Azucares   Grasas_animales          Cereales      Prev.cardiac        Prev.hiper
                0                  0                 0                 0                 0
Mortalidad.adulta               Sexo
                0                  0
```

5.3.42. World Map of diabesity_index



5.5 Códigos

Objective 1:[RPubs - OBJETIVO 1 - PROYECTO II]

Objective 2: [RPubs - OBJETIVO 2 - PROYECTO II]

Objective 3: [RPubs - OBJETIVO 3 - PROYECTO II]

Objective 4: RPubs - OBJETIVO 4 - PROYECTO II

Objective 5: RPubs - OBJETIVO 5 - PROYECTO II

Objective 6: RPubs - OBJETIVO 6 - PROYECTO II

Bibliography

Arredondo, A. (2014). *Type 2 diabetes and health care costs in Latin America: Exploring the need for greater preventive medicine*. *BMC Medicine, 12*, Article 136. https://doi.org/10.1186/s12916-014-0136-z

"López-Calva, L. F., Rodríguez-Chamussy, L., & Székely, M. (2004). *Medición del Desarrollo Humano* en México:Introducción* https://sic.cultura.gob.mx/documentos/1006.pdf